



7.2.3 Observations ¶

Through an example, we have illustrated the following insights regarding the direct solution of sparse linear systems:

- There is a one-to-one correspondence between links in the graph that shows how mesh points are influenced by other mesh points (connectivity) and nonzeros in the matrix. If the graph is undirected, then the sparsity in the matrix is symmetric (provided the unknowns are ordered in the same order as the equations that relate the unknowns to their neighbors). If the graph is directed, then the matrix has a nonsymmetric sparsity pattern.
- Renumbering the mesh points is equivalent to correspondingly permuting the columns of the matrix and the solution vector. Reordering the corresponding equations is equivalent to permuting the rows of the matrix.

These observations relate the problem of reducing fill-in to the problem of partitioning the graph by identifying a separator. The smaller the number of mesh points in the separator (the interface), the smaller the submatrix that corresponds to it and the less fill-in will occur related to this dissection.

Remark 7.2.3.1. Importantly: one can start with a mesh and manipulate it into a matrix or one can start with a matrix and have its sparsity pattern prescribe the graph.